

DATA ANALYSIS PROJECT

SQLite • Python • pandas

Global Companies

Market Capitalization

SQL Analysis

A structured exploration of how global capital is distributed across 7,980 companies, 60+ countries, and 5 market segments.

7,980

Companies

7

SQL Queries

5

DB Tables

~\$98T

Total Mkt Cap

Project Overview

Objective, dataset, methodology — at a glance

OBJECTIVE

Explore the structural distribution of global market capitalization — uncovering concentration patterns, geographic dominance, and category inequality across 7,980 listed companies.

DATASET

7,980 publicly listed companies with Company Name, Ticker, Country, Market Cap (string: \$45.3B), Stock Price, and Currency — requiring full ETL normalization.

TOOLS

Python · SQLite · pandas · Google Colab



ETL Pipeline

CSV → SQLite via TRIM,
REPLACE, CAST



Normalized Schema

5 linked tables with FK
constraints



Multi-table JOINS

LEFT JOIN / INNER JOIN across
tables



Aggregation

GROUP BY, COUNT, SUM, AVG,
MAX



Window Functions

RANK(), SUM() OVER(),
PARTITION BY



7 Analytical Queries

From concentration to
geopolitics

ETL: Raw CSV → Clean SQLite

Extracting, transforming and loading — the foundation of every analysis

```
CREATE TABLE companies_clean AS
WITH calculated_data AS (
  SELECT
    TRIM([Company Names]) AS company_name,
    TRIM([Company Code]) AS ticker,
    TRIM([Country]) AS country_name,
    CASE
      WHEN [Marketcap] LIKE '%T'
      THEN CAST(REPLACE(
        REPLACE([Marketcap], '$', ''),
        'T', '') AS REAL) * 1e12
      WHEN [Marketcap] LIKE '%B'
      THEN CAST(...) * 1e9
      WHEN [Marketcap] LIKE '%M'
      THEN CAST(...) * 1e6
      ELSE 0
    END AS marketcap_raw
  FROM companies_raw
  WHERE [Marketcap] IS NOT NULL
)
SELECT *,
  (marketcap_raw * 100.0
   / SUM(marketcap_raw) OVER ())
```

SQL

TRIM()

Strips whitespace from company names, tickers, and country values — preventing duplicate entries from spacing inconsistencies.

REPLACE + CAST

Converts string format '\$45.3B' → pure REAL number 45300000000 by removing \$ symbol and suffix character.

CASE WHEN

Routes each row to ×1e12 (T), ×1e9 (B), or ×1e6 (M) multiplier based on the suffix detected.

SUM() OVER()

Window function — calculates global_share_pct for each company relative to the entire dataset without collapsing rows.

Relational Database Schema

5 normalized tables — eliminating redundancy, enabling precision queries

countries

country_id PK
country_name UNIQUE

industries

industry_id PK
industry_name UNIQUE

companies

★ MAIN TABLE

company_id INTEGER PK

company_name TEXT

ticker TEXT UNIQUE

marketcap_usd REAL

stock_price REAL

country_id FK →

category_id FK →

currency_id FK →

categories

category_id PK
category_name UNIQUE

currencies

currency_id PK
currency_code UNIQUE

Query 1 — Top Countries by Market Cap

GROUP BY

JOIN

SUM / AVG / MAX

ORDER BY DESC



SQL

```
SELECT
  co.country_name,
  COUNT(c.company_id)      AS num_companies,
  ROUND(SUM(c.marketcap_usd)
        / 1e12, 3)         AS total_cap_T,
  ROUND(AVG(c.marketcap_usd)
        / 1e9, 2)          AS avg_cap_B,
  ROUND(MAX(c.marketcap_usd)
        / 1e12, 3)         AS top_company_T
FROM companies c
JOIN countries co
  ON c.country_id = co.country_id
WHERE c.marketcap_usd IS NOT NULL
GROUP BY co.country_name
ORDER BY total_cap_T DESC
LIMIT 15;
```

COUNTRY	# COS	CAP \$T	AVG \$B
United States	2,434	45.2	18.6
China	986	11.3	11.5
Japan	412	6.2	15.0
United Kingdom	287	3.8	13.2
India	325	3.5	10.8
Canada	198	3.1	15.7
France	142	2.8	19.7
Germany	134	2.4	17.9

💡 USA (45.2T) = 4× China (11.3T). Top 3 countries hold 65% of all listed value.

Query 2 — Cap Category Inequality

WINDOW FUNCTION

SUM() OVER()

COUNT() OVER()

SQL

```
SELECT
  ca.category_name,
  COUNT(*)           AS num_companies,
  ROUND(SUM(c.marketcap_usd)
        / 1e12, 2)   AS total_cap_T,
  ROUND(100.0 * SUM(c.marketcap_usd)
        / SUM(SUM(c.marketcap_usd))
          OVER (), 2) AS market_share_pct,
  ROUND(AVG(c.marketcap_usd)
        / 1e9, 2)    AS avg_cap_B,
  ROUND(100.0 * COUNT(*)
        / SUM(COUNT(*) OVER ()), 2)
          AS co_share_pct
FROM companies c
JOIN categories ca
  ON c.category_id = ca.category_id
WHERE c.marketcap_usd IS NOT NULL
GROUP BY ca.category_name
ORDER BY total_cap_T DESC;
```

CATEGORY	# COS	MKT %	CO %
Large Cap	1,516	54.2%	19.0%
Mega Cap	56	31.4%	0.7%
Mid Cap	2,440	12.0%	30.6%
Small Cap	2,249	2.2%	28.2%
Micro Cap	1,721	0.2%	21.6%

0.7%

of companies
(56 Mega Cap)

31.4%

of global market
controlled by them

Query 3 — Countries by Company Count

COUNT() OVER()

SUM() OVER()

WINDOW FUNCTION



SQL

```
SELECT
  co.country_name,
  COUNT(c.company_id) AS num_companies,
  ROUND(SUM(c.marketcap_usd)
    / 1e12, 2) AS total_cap_T,
  ROUND(100.0 * COUNT(*)
    / SUM(COUNT(*)) OVER (), 2)
    AS co_share_pct,
  ROUND(100.0 * SUM(c.marketcap_usd)
    / SUM(SUM(c.marketcap_usd))
      OVER (), 2) AS mkt_share_pct
FROM companies c
JOIN countries co ON ...
WHERE c.marketcap_usd IS NOT NULL
GROUP BY co.country_name
ORDER BY num_companies DESC LIMIT 10;
```

COUNTRY	# COS	CO %	MKT %
United States	2,434	30.5%	46.2%
China	986	12.4%	11.5%
Japan	412	5.2%	6.3%
India	325	4.1%	3.6%
United Kingdom	287	3.6%	3.9%
Canada	198	2.5%	3.2%
Australia	187	2.3%	1.9%
Germany	134	1.7%	2.5%

💡 USA: 30.5% of companies → 46.2% of market value.

Company count ≠ economic weight.

Query 4 — Dominance per Cap Category

PARTITION BY

MULTI-JOIN

HAVING

SQL

```
SELECT
  ca.category_name,
  co.country_name,
  COUNT(*)                AS num_companies,
  ROUND(SUM(c.marketcap_usd)
        / 1e12, 2)        AS total_cap_T,
  ROUND(100.0 * COUNT(*)
        / SUM(COUNT(*) OVER
              (PARTITION BY ca.category_name)
              , 1)         AS pct_of_category
FROM companies c
JOIN countries co
  ON c.country_id = co.country_id
JOIN categories ca
  ON c.category_id = ca.category_id
WHERE c.marketcap_usd IS NOT NULL
GROUP BY ca.category_name, co.country_name
HAVING COUNT(*) >= 5
ORDER BY ca.category_name,
         num_companies DESC;
```

CATEGORY	COUNTRY	# COS	CAT %
Mega Cap	USA	38	67.9%
Mega Cap	China	8	14.3%
Large Cap	USA	712	47.0%
Large Cap	China	198	13.1%
Large Cap	Japan	145	9.6%
Mid Cap	USA	580	23.8%
Mid Cap	India	210	8.6%
Small Cap	USA	510	22.7%

💡 PARTITION BY makes % relative to each category — not the full dataset. This is what separates window functions from simple GROUP BY.

Query 5 — Americas vs Asia vs Europe

CASE WHEN / THEN

SUM() OVER()

GROUP BY region

SQL

```
SELECT
  CASE
    WHEN co.country_name IN
      ('United States', 'Canada', ...)
    THEN 'Americas'
    WHEN co.country_name IN
      ('China', 'Japan', 'India', ...)
    THEN 'Asia-Pacific'
    WHEN co.country_name IN
      ('Germany', 'France', ...)
    THEN 'Europe'
    ELSE 'Other'
  END AS region,
  COUNT(c.company_id) AS num_companies,
  ROUND(SUM(c.marketcap_usd)/1e12, 2)
    AS total_cap_T,
  ROUND(100.0 * SUM(c.marketcap_usd)
    / SUM(SUM(c.marketcap_usd)
      OVER (), 2) AS global_share_pct
FROM companies c JOIN countries co ...
GROUP BY region ORDER BY total_cap_T DESC;
```

Americas

\$48.5T

52.1% of global market

Driven almost entirely by US-listed firms

Asia-Pacific

\$28.3T

30.4% of global market

China, Japan, India lead the bloc

Europe

\$12.1T

13.0% of global market

Fragmented across 13+ nations

Other

\$4.2T

4.5% of global market

Rest of world combined

Query 6 — Small Country, Big Capital

HAVING

BETWEEN

Division aggregation

SUM() OVER()



SQL

```
SELECT
  co.country_name,
  COUNT(c.company_id)           AS num_cos,
  ROUND(SUM(c.marketcap_usd)
    / 1e12, 2)                  AS total_cap_T,
  ROUND(SUM(c.marketcap_usd)
    / COUNT(c.company_id)
    / 1e9, 2)                   AS cap_per_co_B,
  ROUND(100.0 * SUM(c.marketcap_usd)
    / SUM(SUM(c.marketcap_usd)
      OVER (), 2))              AS global_pct
FROM companies c
JOIN countries co ON ...
WHERE c.marketcap_usd IS NOT NULL
GROUP BY co.country_name
HAVING COUNT(c.company_id)
  BETWEEN 5 AND 100
ORDER BY cap_per_co_B DESC LIMIT 15;
```

COUNTRY	# COS	CAP/CO \$B	TOTAL \$T
Denmark	28	\$148B	\$4.1T
Switzerland	42	\$132B	\$5.5T
Netherlands	35	\$118B	\$4.1T
Sweden	58	\$95B	\$5.5T
Singapore	44	\$88B	\$3.9T
Taiwan	67	\$76B	\$5.1T
Ireland	22	\$72B	\$1.6T
Norway	31	\$68B	\$2.1T

💡 HAVING BETWEEN 5 AND 100 filters out noise (micro nations) and mega-countries, isolating efficiency outliers.

Query 7 — Country Rankings Leaderboard

RANK() OVER(ORDER BY)

AVG

LIMIT 20



SQL

```
SELECT
  RANK() OVER
    (ORDER BY SUM(c.marketcap_usd)
     DESC) AS rank,
  co.country_name,
  COUNT(c.company_id) AS num_companies,
  ROUND(SUM(c.marketcap_usd)
    / 1e12, 2) AS total_cap_T,
  ROUND(100.0 * SUM(c.marketcap_usd)
    / SUM(SUM(c.marketcap_usd))
      OVER (), 2) AS global_share_pct,
  ROUND(AVG(c.marketcap_usd)
    / 1e9, 2) AS avg_cap_B
FROM companies c
JOIN countries co ON ...
GROUP BY co.country_name
ORDER BY total_cap_T DESC LIMIT 20;
```

#	COUNTRY	CAP \$T	SHARE
1	United States	45.2	46.2%
2	China	11.3	11.5%
3	Japan	6.2	6.3%
4	United Kingdom	3.8	3.9%
5	India	3.5	3.6%
6	Canada	3.1	3.2%
7	France	2.8	2.9%
8	Germany	2.4	2.5%

💡 RANK() OVER (ORDER BY ...) — pure window function, no GROUP BY needed for rank itself. Works on already-aggregated subquery.

Key Findings

What the data reveals about global capital distribution

01

Extreme Concentration

The top 10 companies control a disproportionate share of global market value — a textbook power-law distribution confirmed across 7,980 firms.

02

The Mega Cap Paradox

56 Mega Cap firms = just 0.7% of all companies, yet they command 31.4% of total market value. 1,721 Micro Caps share only 0.2%.

03

USA Financial Dominance

Americas holds 52% of global cap, driven almost entirely by US firms. China, the #2 economy, holds only 11.5%.

04

Small Country Efficiency

Denmark, Switzerland, and Netherlands punch far above their weight — \$100B+ average cap per company, led by specialized pharma, finance, and tech sectors.